# FIPS 140-3 (Cryptographic Module Validation)

**Anticloud FZ LLE — PAX L5 Narrow L2 General 27B**

Run: 2026-10-02T11:18:03.843415Z | Framework ID: 14_FIPS_140_3

## Standard

FIPS 140-3 (NIST, 2019), ISO/IEC 19790:2012

## Result

PARTIAL — SHA3-256 (FIPS-approved), AES-256-GCM (FIPS-approved); scrypt key derivation not FIPS-approved

## Score

Level 2 target — SHA3/AES approved; scrypt requires substitution with PBKDF2/HKDF for full FIPS

## Notes

SHA3-256 (FIPS 202), AES-256-GCM (SP 800-38D) are FIPS-approved. Recommendation: replace scrypt with PBKDF2-SHA512.

## Real Benchmark Numbers (Kaggle T4, loiskleinner/pax-benchmark-runner v3)

| Module | Key Metric | Value |
|---|---|---|
| CRDT | lww_set_us | 0.51 |
| CRDT | merge_10k_ms | 0.07 |
| CRDT | convergence | GUARANTEED |
| CryptographicProvenance | write_per_receipt_ms | 0.059 |
| CryptographicProvenance | verify_1k_chain_ms | 2.53 |
| ZeroTrustLocalMesh | ca_gen_ms | 367.5 |
| ZeroTrustLocalMesh | cert_issue_ms | 333.8 |
| ZeroTrustLocalMesh | tls_version | 1.3 |
| LocalObservabilityStack | spans_per_sec | 17558 |
| LocalObservabilityStack | logs_per_sec | 24492 |
| SovereignMemory | write_per_entry_ms | 51.94 |
| SovereignMemory | encryption | AES-256-GCM |
| AIOSS_AuditChain | entries_per_sec | 232665 |

| AIOSS_AuditChain | hash_algo | SHA3-256 |

## Environment Provenance (Preprioception)

| Field | Value |
|---|---|
| Run ID | loiskleinner/pax-benchmark-runner v3 |
| Run Date | 2026-10-02 |
| Platform | Kaggle T4 x2 GPU, Ubuntu 22.04, Python 3.12 |
| GPU | NVIDIA Tesla T4 (2× 16 GB VRAM) |
| CPU | Intel Xeon (2 vCPU, 13 GB RAM) |
| Inference Temperature | N/A (code benchmarks, no LLM sampling) |
| KV Cache | N/A (no LLM inference in this cell) |
| Framework | llama-cpp-python 0.3.x (installed from source with CUDA) |

## Cryptographic Integrity

| Field | Value |
|---|---|
| SHA3-256 (content) | 16055c2d7559282dd803f09a0894518dcc392004198fc069d25930d28cda702d |
| SHA3-256 (chain) | c9c6dfbb9a3025be08592e811539949a5bd8aec99060aec8b918b85efeb11bdd |
| Timestamp UTC | 2026-10-02T11:18:03.843415Z |
| Hash Algorithm | SHA3-256 (FIPS 202 compliant) |

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*Anticloud FZ LLE · PAX L5 Narrow L2 General 27B · AIOSS Ledger Genesis: 8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560*